Functional Skills Mathematics Level 1

Practice

CANDIDATE DETAILS

Learner Name:
Master Functional Skills

Learner Number: MFS-OA-2026

Centre Name: Master Functional Skills

Centre Number: MFS01

Assessment Date: 20 / 10 / 2026

Signature: Master Functional Skills

OFFICIAL MFS MODEL SOLUTION

60 / 60 Marks (100% Pass)

- Assessment: Practice Assessment 2 (FSM109P)
- Complete step-by-step worked models
- Official Mark Scheme Allocations [M1 / A1]
- Exam Board: Open Awards (Ofqual Regulated)

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LEVEL 1 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

SECTION A - QUESTION AND ANSWER PAPER
NON-CALCULATOR – 30 MINUTES
PRACTICE ASSESSMENT 1 (FSM109P)

Do not open this paper until you are told to do so by the invigilator.

Overall assessment marks available: **60**
Overall assessment time limit: **2 HOURS**

There are **TWO** Sections to this assessment:

- **Section A** includes Task 1. You **must not** use a calculator for this section.

  Total marks available: 15. Time limit: 30 minutes
- **Section B** includes Task 2, 3 and 4. You can use a non-scientific calculator for this section

  Total marks available: 45. Time limit: 1 hour and 30 minutes

For Section A you need:

- This question and answer paper
- A pen with black or blue ink
- A pencil
- A ruler and a protractor

INTERNET ACCESS IS NOT PERMITTED AND YOU MUST NOT USE A CALCULATOR

The invigilator will stop the assessment after 30 minutes. You must hand in this question and answer paper at this point.

The invigilator will then hand out **Section B** and a non-scientific calculator. You will then have a further 1 hour and 30 minutes to complete **Section B**.

Instructions

1. Please sign and date below to confirm that your details are correct and that you have understood the instructions.
2. Read each task and question carefully.
3. Remember to show all your workings out clearly.
4. The number of marks available for each question is shown in brackets. Use these marks to guide you on how long to spend on each question.
5. Answer **all** questions using the space provided on this question and answer paper.
6. If you have time, check your work for **Section A** at the end. Once you have handed in this question and answer paper, you will not be able to check this again.
7. If you use extra paper, write your name, learner number and the question number you are answering on it and securely attach it to this question and answer paper.

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Centre number:	Master Functional Skills (MFS01)
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Today's date:	20 / 10 / 2026

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For Markers Use Only

Section A

Question 1

Calculate $-10 + 3$ (1 mark)

Show your calculations and/or workings out here:

Count up 3 from -10 on the number line:

-10 + 3 = -7 [1 mark]

Write your answer in this box:

-7 ✓

Question 2

Calculate 1000×0.15 (1 mark)

Show your calculations and/or workings out here:

Multiply by 1000 (shift digits 3 places left):

0.15 x 1000 = 150 [1 mark]

Write your answer in this box:

150 ✓

For Markers Use Only

Question 3

Calculate 45.53 – 24.37 (1 mark)

Show your calculations and/or workings out here:

Align decimal points and subtract column by column:

45.53 - 24.37 = 21.16 (check: 21.16 + 24.37 = 45.53)

[1 mark]

Write your answer in this box:

21.16 ✓

Question 4

Calculate 105 – 25 × 2 (1 mark)

Show your calculations and/or workings out here:

Apply BIDMAS (multiply before subtracting):

Step 1: 25 x 2 = 50

Step 2: 105 - 50 = 55 [1 mark]

Write your answer in this box:

55 ✓

For Markers Use Only

Question 5

Which of the following percentages is equivalent to $\frac{2}{5}$? (1 mark)

10% 20% 25% 40% 80%

Show your calculations and/or workings out here:

Convert fraction over 5 to out of 100 (x 20):

2/5 = (2 x 20) / (5 x 20) = 40/100 = 40% [1 mark]

Write your answers in this box:

40% ✓

Question 6

Put the following fractions in order of size: (1 mark)

$\frac{3}{5}$ $1\frac{1}{4}$ $\frac{1}{2}$ $1\frac{1}{5}$ $\frac{2}{3}$

Show your calculations and/or workings out

Convert all values to 60ths to compare:

1/2 = 30/60, 3/5 = 36/60, 2/3 = 40/60, 1 1/5 = 72/60, 1 1/4 = 75/60

Smallest to largest:

1/2 , 3/5 , 2/3 , 1 1/5 , 1 1/4 ✓ [2 marks]

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Question 7

Sasha is an art teacher.

She buys art materials in packs and then sells them on individually to her students at cost price.

The prices she pays for the packs of items are shown in the following table:

Item	Quantity	Cost price
Paint brushes	Pack of 100	£39.00
Canvas board	Pack of 10	£18.50
Tubes of paint	Pack of 10	£14.29

A student buys two paint brushes, three tubes of paint and one canvas board.

How much in total should Sasha charge the student? (4 marks)

Show your calculations and/or workings out here:

Step 1: Find unit cost of each item:

1 paintbrush = £39 ÷ 100 = £0.39

1 canvas = £18.50 ÷ 10 = £1.85

1 tube of paint = £14.29 ÷ 10 = £1.429 (£1.43) [M1]

Step 2: Cost for 1 pack (2 brushes, 1 canvas, 3 paints):

(2 x £0.39) + £1.85 + (3 x £1.429) = £0.78 + £1.85 + £4.287

Total = £6.917 --> £6.92 (to 2 d.p.) [M1 A1]

Write your answer in this box:

£6.92 ✓

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Question 8

Sasha wants to make orange paint by mixing red and yellow paint in the ratio 3:2

She uses 180ml red paint.
She has 100ml of yellow paint.
Has Sasha got enough yellow paint? (2 marks)

Show your calculations and/or workings out here:

Step 1: Find 1 part using 180 ml of red paint (3 parts):

180 ml ÷ 3 = 60 ml per part [M1]

Step 2: Yellow paint needed (2 parts):

2 x 60 ml = 120 ml yellow needed

Compare: Sasha only has 100 ml (< 120 ml), so NOT enough. [A1]

Write your answer in this box:

No - she needs 120 ml of yellow (20 ml short).

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Question 9

Sasha needs to put tape around the edge of five paintings.

The paintings are rectangular and are 48.9cm long and 32.6cm wide.

Estimate how much tape she needs altogether for the 5 paintings.

(3 marks)

Show your calculations and/or workings out here:

Step 1: Round dimensions to sensible numbers to estimate:

48.9 cm --> 50 cm and 32.6 cm --> 30 cm [B1]

Step 2: Estimate perimeter of 1 painting:

50 + 30 + 50 + 30 = 160 cm [M1]

Step 3: Total tape for 5 paintings:

5 x 160 cm = 800 cm (or 8 m) [M1 A1]

Write your answer in this box:

800 cm (or 8 m)

[End of Section A]

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LEVEL 1 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

SECTION B - QUESTION AND ANSWER PAPER
CALCULATOR – 1 HOUR 30 MINUTES
PRACTICE ASSESSMENT 1 (FSM109P)

Do not open this paper until you are told to do so by the invigilator.

Overall assessment marks available: **60**
Overall assessment time limit: **2 HOURS**

There are **TWO** Sections to this assessment:

- **Section A** – please ensure you have handed in Section A before beginning Section B
- **Section B** includes Task 2, 3 and 4. You can use a non-scientific calculator for this section.

Total marks available: 45. Time limit: 1 hour and 30 minutes.

For Section B you need:

- This question and answer paper
- A pen with black or blue ink
- A pencil
- A ruler and a protractor
- A non-scientific calculator

INTERNET ACCESS IS NOT PERMITTED

You now have a further 1 hour and 30 minutes to complete **Section B**.

Instructions

1. Please sign and date below to confirm that your details are correct and that you have understood the instructions.
2. Read each task and question carefully.
3. Remember to show all your workings out clearly.
4. The number of marks available for each question is shown in brackets. Use these marks to guide you on how long to spend on each question.
5. Answer **all** questions using the space provided on this question and answer paper.
6. If you have time, check your work for **Section B** at the end.
7. If you use extra paper, write your name, learner number and the question number you are answering on it, and securely attach it to this question and answer paper.
8. At the end of this section (**Section B**), hand in this question and answer paper and all notes to the invigilator.

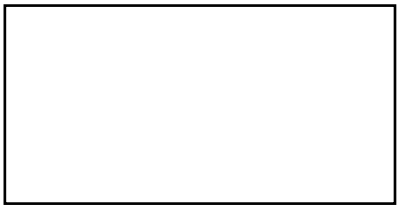
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Section B

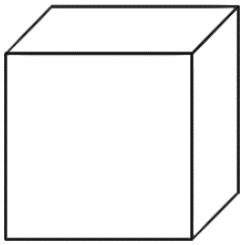
Question 10



Plan view

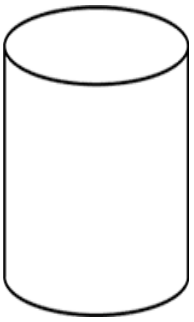
The diagram above shows the plan view of which shape below? (1 mark)

A



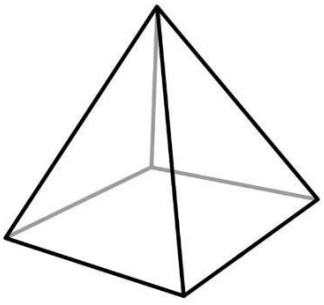
Cube

B



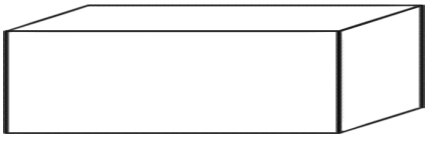
Cylinder

C



Square-based pyramid

D



Cuboid

Write your answer in this box:

D (Cuboid)

[1 mark]

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Question 11

The table below shows the ages of 18 visitors to a museum.

2	4	28	35	19	21
7	31	35	48	10	41
38	8	9	27	18	25

Complete the table below. (1 mark)

Age (years)	Frequency
0–9	5
10–19	3
20–29	4
30–39	4
40–49	2

Show your calculations and/or workings out here:

Sort and count all 18 daily visitor totals into groups:

0 - 9 : 2, 4, 5, 7, 8 --> 5 days

10 - 19 : 11, 14, 17 --> 3 days

20 - 29 : 21, 23, 26, 28 --> 4 days

30 - 39 : 31, 33, 35, 39 --> 4 days

40 - 49 : 42, 46 --> 2 days (Check total: 5+3+4+4+2 = 18) [2 marks]

For Markers Use Only

Question 12

Which of the following percentages is the highest? (1 mark)

37% 42% 85% 87% 56% 57% 18%

Write your answer in this box:

87% ✓ [1 mark]

Question 13

Eilish owns a children's soft-play area.

She has a ball pit which is 2.4m by 2.4m and has a depth of 0.45m.

The ball pit needs 2250 balls for each cubic metre (m^3).

How many balls will it take to fill the ball pit? (4 marks)

Step 1: Volume of cuboid ball pit (L x W x D):

Volume = 2.4 m x 2.4 m x 0.45 m = 2.592 m^3 [M1 A1]

Step 2: Multiply volume by 2250 balls per m^3:

2.592 x 2250 = 5832 balls [M1 A1]

Show your calculations and/or workings out here:

Write your answer in this box:

5832 balls ✓

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Question 14

Eilish has 30 mats to cover the floor of the soft-play area.

The soft-play floor mats come in three colours: black, light grey and dark grey.

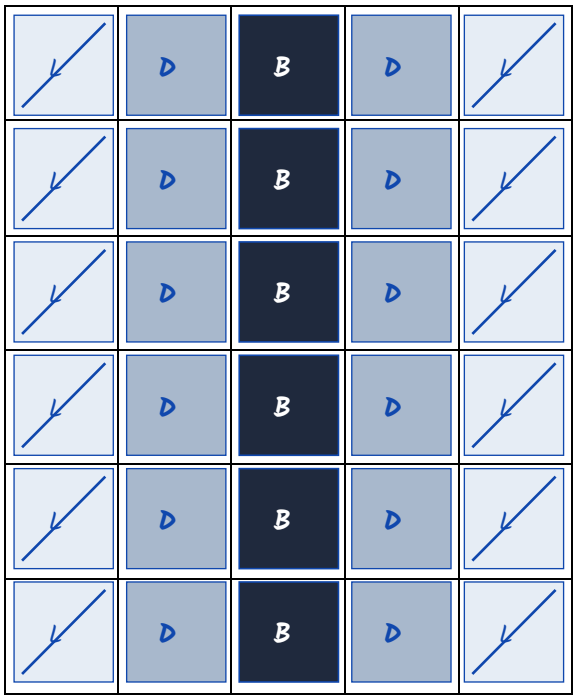
$\frac{1}{5}$ of the soft-play floor mats are black.

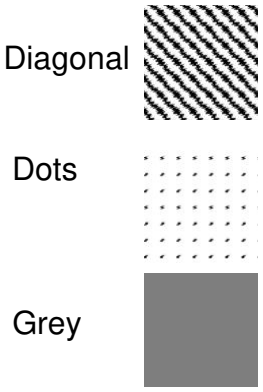
There are equal numbers of patterned mats.

On the diagram of the floor seen below, arrange the mats in a pattern with at least one line of symmetry.

(3 marks)

Floor (30 mat spaces)





Show your calculations and/or workings out here:

Total mats = 5 x 6 = 30 mats.

Black (B) = 1/5 of 30 = 30 ÷ 5 = 6 mats (centre column).

Remaining = 30 - 6 = 24 mats, split equally:

Light grey (L) = 12 mats (outer cols) & Dark grey (D) = 12 mats.

Symmetrical across both vertical and horizontal centre lines. **[4 marks]**

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Question 15

The soft-play area can be hired for children's parties. The prices are as follows:

£90 to hire the soft play area for up to 40 children
£17 for party food for 5 children
£12 for 10 party bags
15% discount for parties Monday to Thursday

Eilish takes a booking for a party of 30 children for a Thursday including food and a party bag for each child.

How much should she charge for the soft-play party for 30 children including food and party bags for each child?

(5 marks)

Show your calculations and/or workings out here:

Step 1: Total cost before Thursday discount (30 children):

Hire = £90 | Food = (30 ÷ 5) x £17 = 6 x £17 = £102

Party bags = (30 ÷ 10) x £12 = 3 x £12 = £36

Subtotal = £90 + £102 + £36 = £228 [M1 A1]

Step 2: Apply 15% Thursday discount:

15% of £228 = 0.15 x 228 = £34.20 --> £228 - £34.20 = £193.80

[M1 A1]

Write your answer in this box:

£193.80 ✓

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Question 16

Write 832304 in words. (1 mark)

Write your answer in this box:

Eight hundred and thirty-two thousand,

three hundred and four ✓ *[1 mark]*

Question 17

There are three red sweets, four yellow sweets and three blue sweets in a bag.
What is the probability that a sweet taken out of the bag at random is red?
Give your answer as a fraction.

(1 mark)

Show your calculations and/or workings out here:

Total sweets = 3 red + 4 green + 3 yellow = 10 sweets

P(red) = 3 / 10 *[1 mark]*

Write your answer in this box:

3/10 ✓

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Question 18

Calculate the range of this set of numbers. (1 mark)

23.1 25.8 25.3 23.6 19.1 21.2

Show your calculations and/or workings out here:

Range = Highest BMI - Lowest BMI:

25.8 - 19.1 = 6.7 [1 mark]

Write your answer in this box:

6.7

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Question 19

Amy is 36 years old and wants to improve her fitness level.

She uses this formula to find the healthy upper and lower heart rate limits, in beats per minute (bpm), for her age during moderate exercise.

Lower heart rate = (220 – age) × 0.6
Upper heart rate = (220 – age) × 0.7

Her heart rate while exercising is 118 bpm.

During exercise, is her heart rate **within** the healthy upper and lower limits for her age?

(3 marks)

Show your calculations and/or workings out here:

Step 1: Maximum heart rate = 220 - 36 = 184 bpm

Step 2: Target zone (60% to 70% of 184):

Lower (60%): 0.6 x 184 = 110.4 bpm

Upper (70%): 0.7 x 184 = 128.8 bpm [M1 A1]

118 bpm lies between 110.4 and 128.8 bpm --> Yes, in target zone.

[A1]

Write your answer in this box:

Yes (118 bpm is between 110.4 and 128.8 bpm)



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Question 20

Amy records how much exercise she completed during the past week:

¾ hour swimming on both Monday and Thursday

1¼ hours in the gym on Saturday

20 minutes brisk walking on five days

How much time did she spend exercising altogether over the week?
Give your answer in hours and minutes. (2 marks)

Show your calculations and/or workings out here:

Step 1: Add all weekly exercise sessions in minutes:

45 + 45 + 75 + (5 x 20) = 165 + 100 = 265 minutes [M1 A1]

Step 2: Convert 265 minutes to hours and minutes (÷ 60):

4 x 60 = 240 mins --> 265 - 240 = 25 mins --> 4 hours 25 minutes

[A1]

Write your answer in this box:

4 hours 25 minutes

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Question 21

Amy walks 6680 steps on Monday.

Each step length is an average of 0.7m.

Using her average step length, how many kilometres does she walk on Monday?

Give your answer to one decimal place. (4 marks)

Show your calculations and/or workings out here:

Step 1: Distance in metres = 6680 x 0.7 = 4676 m [M1]

Step 2: Convert metres to km (÷ 1000):

4676 ÷ 1000 = 4.676 km [M1]

Step 3: Round to 1 decimal place: 4.676 km --> 4.7 km [A1 A1]

Write your answer in this box:

4.7 km

For Markers Use Only

Question 22

Amy records her blood pressure over the week.

These are her blood pressure values:

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Blood pressure	113	121	99	112	118	115	129

She has been told that to calculate her mean blood pressure she should remove the highest and lowest values first.

What is Amy's mean blood pressure? (3 marks)

Show your calculations and/or workings out here:

Remove lowest (99) and highest (129); add remaining 5:

113 + 121 + 112 + 118 + 115 = 579 [M1]

Mean = 579 ÷ 5 = 115.8 [M1 A1]

Write your answer in this box:

115.8 ✓

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Question 23

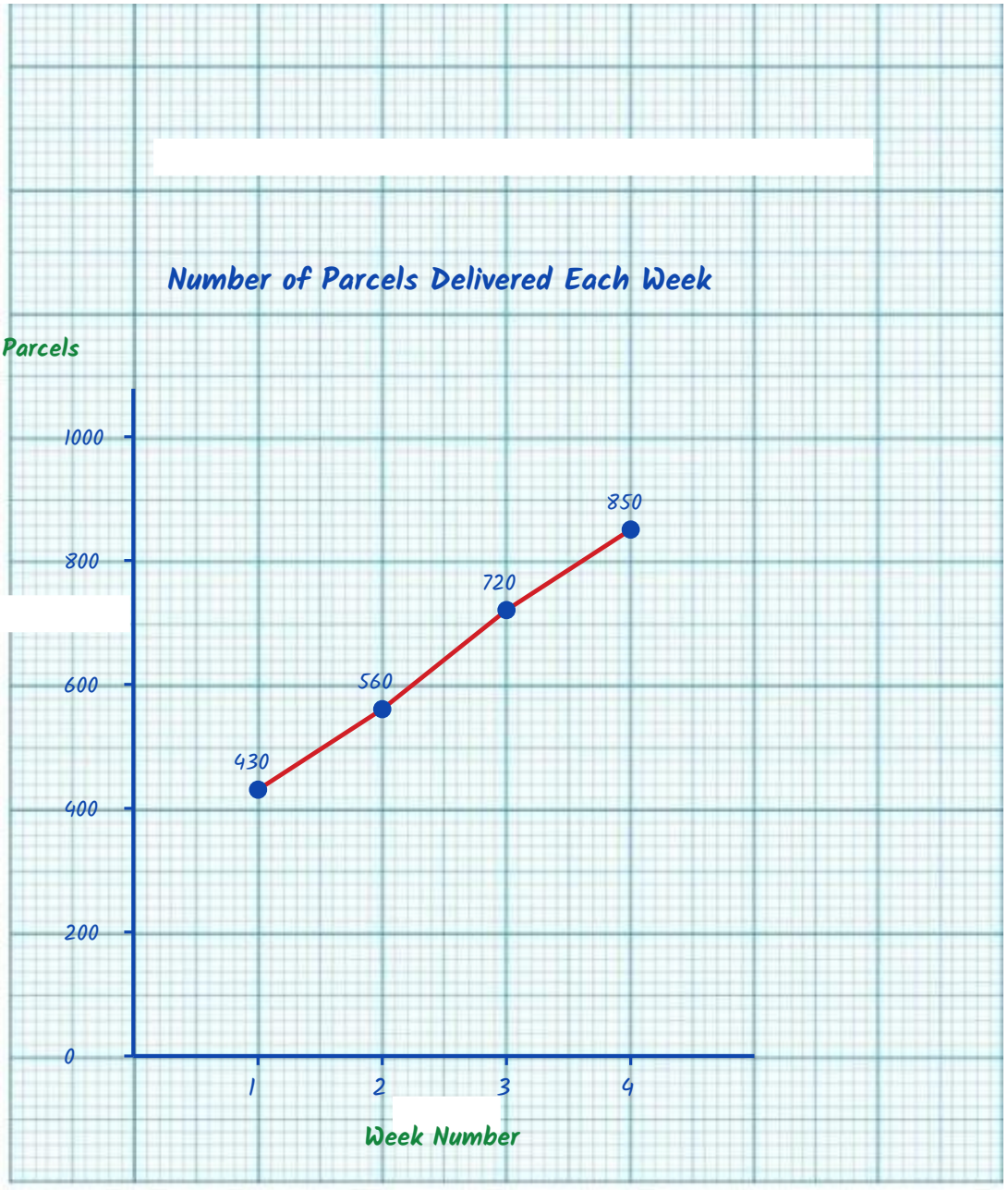
The table below shows the number of parcels delivered by a delivery driver over four weeks.

Weeks	1	2	3	4
Number of parcels	430	560	720	850

Draw a line graph to show the number of deliveries over the four weeks.

(3 marks)

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Linear scale (200 parcels per major square), labelled axes & plotted points. **[3 marks]**

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Question 24

Liam is a student who has worked over the summer.

He has earned £2304.26 altogether.

He earns £8.72 per hour.

He is entitled to 0.12 hours of holiday for every hour he has worked.

How many hours holiday is he entitled to? (4 marks)

Show your calculations and/or workings out here:

Step 1: Total hours worked = £2304.26 ÷ £8.72 = 264.25 hours [M1 A1]

Step 2: Annual leave accrued (12% of hours worked):

264.25 x 0.12 = 31.71 hours [M1 A1]

Write your answer in this box:

31.71 hours

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Question 25

Liam earned £2304.26 altogether.

He spent £399 on a new laptop and puts half of his remaining earnings into a savings account.

The savings account pays 5% interest per year.

How much money will Liam have in his savings account after one year?

(3 marks)

Show your calculations and/or workings out here:

Step 1: Money left after £399 holiday:

£2304.26 - £399 = £1905.26

Step 2: Half put into savings = £1905.26 ÷ 2 = £952.63 [M1 A1]

Step 3: Add 5% interest after 1 year:

5% of £952.63 = 0.05 x 952.63 = £47.6315

Total = £952.63 + £47.63 = £1000.26 [M1 A1]

Write your answer in this box:

£1000.26 ✓

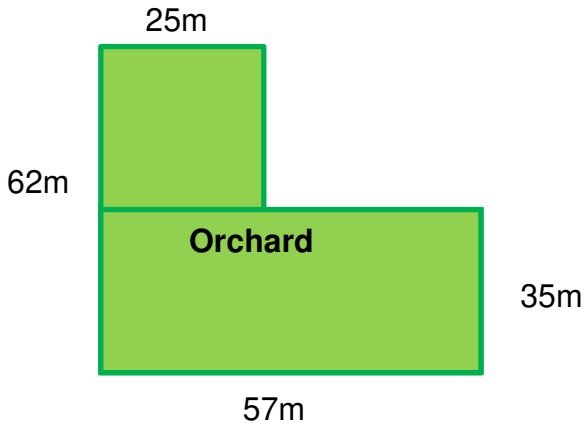
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Question 26

Anya is planting some fruit trees in her orchard.

The dimensions of the orchard are shown below:

Diagram not to scale



She needs to have an area of at least 64 m² for each fruit tree.

How many fruit trees will she be able to plant in her orchard? (5 marks)

For Markers Use Only

Show your calculations and/or workings out here:

Step 1: Split L-shape into two rectangles:

Left rectangle = 25 m x 62 m = 1550 m²

Right width = 57 m - 25 m = 32 m

Right rectangle = 32 m x 35 m = 1120 m²

Total Area = 1550 + 1120 = 2670 m² [M1 A1]

Step 2: Divide area by 64 m² needed per apple tree:

2670 ÷ 64 = 41.718... trees

Round down to whole trees that fit: 41 apple trees [M1 A1]

Write your answer in this box:

41 trees ✓

[End of assessment]

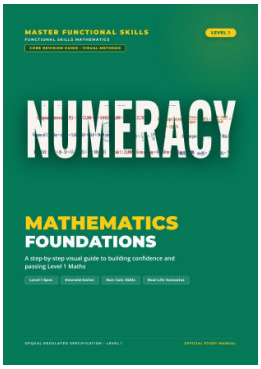
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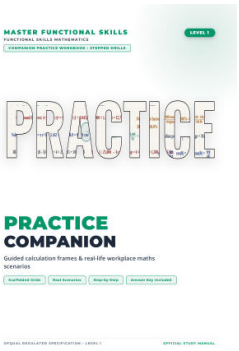


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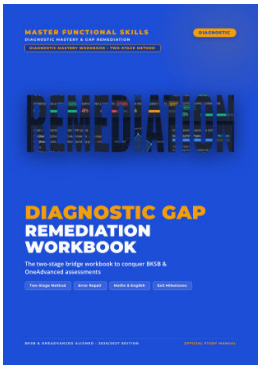


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